

THE CHALLENGES OF BASEMENT INSULATION

A study explored how effective interior insulation systems are at keeping basements dry as well as warm.

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In 2001, Cambridge Homes—a division of D.R. Horton in the Chicago area—completed the initial phase of its first Energy Star development. One of the builder's major improvements of these homes was the insulation of the basement walls. While the homes met the Energy Star goals, problems arose during the first summer the homes were occupied. During a hot, humid period in August, several homeowners reported wet, dripping fiberglass insulation at the foot of their basement walls. Where this moisture came from (humid basement air condensing on the cold wall, drying concrete, groundwater or rain, or some combination of these factors) was unclear, but it certainly needed a remedy. One homeowner was so concerned that she called the EPA to report the problem.

The builder knew that to reach Energy Star levels for homes, basement insulation was key. The insulation system that Cambridge employed was polymer-faced, R-11 fiberglass blankets draped inside the entire basement wall. The builder's immediate solution to the moisture problems was to trim all of the fiberglass blankets 6–12 inches above the floor—eliminating the place where the moisture collected. This action did seem to solve the immediate problems, but it was clear that a better understanding of good basement insulation practices was needed.

Until fairly recently, addressing heat loss from basements has taken a back seat to addressing losses from above-grade



The Chicago basement has four kinds of insulation systems, left to right: polyisocyanurate, two with encapsulated batts in frame walls, and fiberglass blankets.

walls, windows, and ceilings. As above-grade envelopes have improved, foundation heat losses have become an increasing fraction of total home heat load. Exterior foundation wall insulation (generally rigid foam) is often desirable for thermal performance, but protecting this insulation at and above grade can be challenging. For this reason, many builders choose to install insulation on the inside of basement walls. But insulating basements on the inside, as the example above shows, is not always straightforward (see “Dry Notes from the Underground,” *HE*, Mar/Apr '02, p. 18).

Through DOE's Building America program, Steven Winter Associates, Incorporated (SWA), where the authors work, has been working with Cambridge Homes to help improve the energy performance of its houses. In the fall of

2002, Cambridge completed a model home near Chicago. The Consortium for Advanced Residential Buildings (CARB), a team of designers, home builders, and product manufacturers formed by SWA to work with the Building America program, was able to use this home for insulation research. The goal of the research was to assess the thermal and moisture performance of eight production-friendly, interior basement insulation strategies. The findings suggest that rigid foam attached directly to foundation walls is one of the best-performing systems.

The Insulation Systems Tested

Eight insulation systems were installed in this model home: two fiberglass blanket systems, two rigid foam systems, and

four framed-wall systems. Each test section of the basement wall was at least 12 ft wide. SWA meticulously air sealed between all the sections with framing, tape, and foam to ensure that air and moisture transport between adjacent systems was minimized.

In each wall system, temperature and humidity sensors were installed at three vertical locations: 1 ft above the slab floor, 1 ft below the top of the concrete wall, and in the center of the 100-inch wall (see Figure 1). Most wall systems contained two such vertical profiles: one set of sensors against the concrete wall and one set near the inside of the wall system. One system had only one profile (polyisocyanurate) and one had three profiles (kraft-faced batts in the finished wall). Temperature and relative humidity sensors were also installed outdoors and near the center of the basement. Data from all sensors were recorded using a Campbell Scientific datalogger system.

System 1: Draped Fiberglass Blanket with Unperforated Vapor Barrier

When Cambridge insulates an unfinished basement, its current practice is to drape 4-ft-wide, polymer-faced fiberglass blankets vertically down the basement wall. The polymer faces the interior of the basement, and its edges are attached to vertical 2 x 2 furring strips nailed into the concrete wall. At the floor, the bottom 6 inches of concrete are left exposed to allow for moisture dissipation. SWA mimicked this system in the test using an R-11 fiberglass blanket (see Figure 2).

System 2: Draped Fiberglass Blanket with Perforated Facing

In installation and composition, this system is very simi-

lar to system 1; the significant difference is that this fiberglass product is faced with a perforated polymer film. All portions of the basement that do not contain an instrumented insulation system are insulated with this system.

System 3: Kraft-Faced Batts in a Finished Wall

When a Cambridge customer selects a finished-basement option, the standard method for insulating the basement is R-11 kraft-faced batts in 2 x 4 framed walls with studs 16 inches on center. To correct for irregularities in the concrete—and because pressure-treated wood is necessary for direct contact with concrete—framing is located approximately 1 inch inside the foundation wall. The kraft paper is stapled to the faces of studs.

Systems 4 and 5: Encapsulated Fiberglass Batts

Encapsulated fiberglass batts (from two different manufacturers) are installed in 2 x 4 framed walls with studs 16 inches on center. The batts have no perforations on the inside surface and many small perforations on the outside (facing the concrete). The stud wall was located approximately 1 inch from the foundation wall. Polymer tabs were stapled to the stud faces, and the seams between the segments of insulation are taped with Tyvek tape.

System 6: Unfaced Fiberglass Batts

The final framed-wall system consists of a finished,

Location of Sensors in Wall Systems

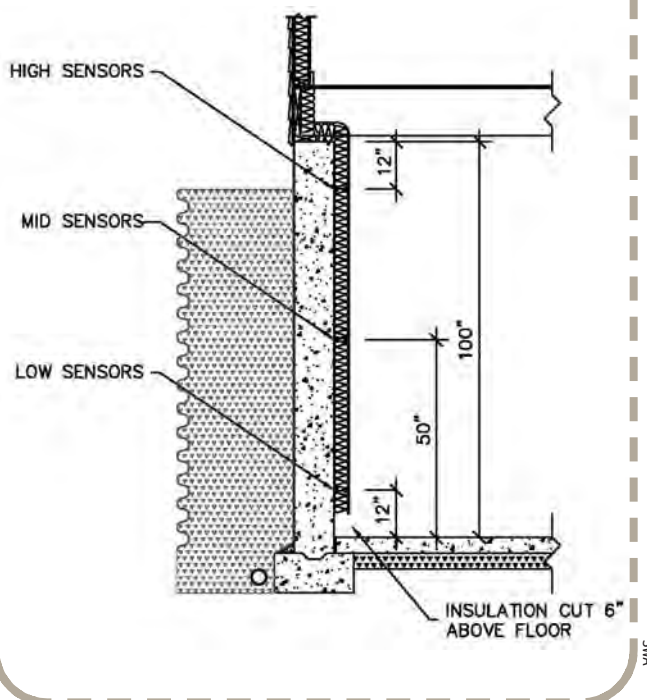


Figure 1. In each wall system, temperature and humidity sensors were installed at three vertical locations: 1 ft above the slab floor, 1 ft below the top of the concrete wall, and in the center of the 100-inch wall.

Polymer-Faced, Fiberglass Blanket Insulation Detail

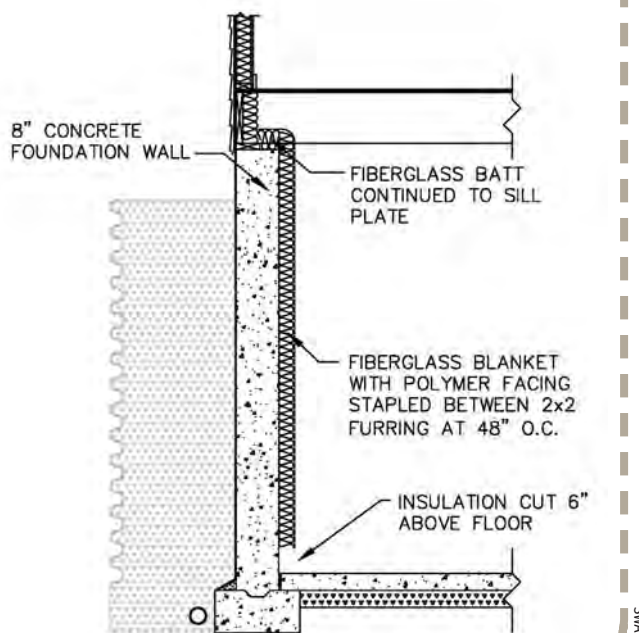


Figure 2. When Cambridge insulates an unfinished basement, its current practice is to drape 4-ft-wide, polymer-faced fiberglass blankets vertically down the basement wall.

framed wall with 2 x 4 studs at 24 inches on center. Between the studs are R-11 friction-fit fiberglass batts. The wall was then dry-walled and painted.

System 7: Polyisocyanurate

Foil-faced, 1 1/2-inch polyisocyanurate boards are attached directly to the concrete, using nails driven through 1 x 4 furring strips. Foil tape is used to cover seams in the polyiso and to cover the exposed edges.

System 8: Expanded Polystyrene

The final section of the basement wall was insulated with 2 inches of expanded polystyrene (EPS) attached to the wall beneath 1 x 4 furring strips. The wall was then drywalled and painted.

The Roads to Failure

With interior basement insulation systems, there are three factors that can cause moisture failure:

- moist indoor air moving through or behind the insulation and condensing on the cool foundation wall;
- moisture from the basement diffusing through the insulation system and condensing on the cool foundation wall; and
- exterior moisture (groundwater, rain) penetrating the foundation wall and getting trapped behind the insulation system.

Each insulation system tested proved to have strengths and weaknesses with respect to each of these factors.

Air Movement

The first surprise from the research was that all four framed-wall systems with fiberglass batts performed almost identically with respect to moisture. Air behind the framed walls had a moisture content (absolute humidity) equivalent to that of the basement air itself. Hour by hour, day by day, season by season, the absolute humidity of the air behind the insulation tracked the humidity of the air in the basement. SWA surmised

that bulk air movement was responsible for this.

A common builder detail for insulating the top of framed basement walls is to lay sections of fiberglass batt across the top of the frame and the foundation wall (see Figure 3). This detail was used in SWA's test walls. Another all too common builder detail is not to insulate the top of the basement wall at all.

In the Chicago basement, encapsulated fiberglass batts were installed in two separate side-by-side wall sections. To test the theory that air movement was responsible for moisture movement, one of these sections was meticulously sealed from basement air with rigid foam, expanding foam, and appropriate tapes. This level of air sealing is



To correct for air gaps behind some fiberglass blanket insulation, SWA modified the insulation manufacturer's installation detail and installed 2 x 2 horizontal furring strips at the top and bottom of each wall section.

much more meticulous than can be expected routinely from builders or insulation contractors. After sealing, moisture content behind the sealed section did indeed stop directly tracking that of the basement air.

During humid summer weather, readings from most framed, fiberglass wall sections showed condensing conditions. The one air sealed section, however, was buffered from excessive basement humidity and did not show signs of condensation. It is also worth noting that during the winter, temperatures behind the sealed fiberglass wall were significantly lower than temperatures behind the unsealed wall. While SWA did not install sensors to compare heat transfer accurately, the temperature difference suggested a 10% reduction in

effective R-value of the unsealed wall section when compared to the sealed section.

Originally, fiberglass blanket systems appeared to perform similarly to the framed fiberglass batt walls: Absolute humidity behind the facing was often similar to that of the basement air. Examination of the blanket installation revealed that there was indeed a substantial air gap behind many of the 4-ft-wide blankets through which basement air could flow. To correct this, SWA modified the insulation manufacturer's installation detail and installed 2 x 2 horizontal furring strips at the top and bottom of each wall section. As with the framed wall sections, the installation details for fiberglass blankets are more important than the materials themselves, because of the potential for air movement.

As expected, the rigid foam systems adhered directly to the foundation walls do not allow air to move behind or through them. Both the EPS and polyisocyanurate sections performed well with respect to air movement.

Moisture Diffusion

Many of the previous studies on interior basement insulation have focused on moisture diffusion and permeability of wall materials. SWA's project found that addressing moisture diffusion is drastically less important than addressing air movement in wall systems. That being said, moisture diffusion can be useful in allowing foundation walls to dry (both when concrete walls are new and when moisture intrudes from an outside source). Diffusion can also be detrimental if moisture migrates from the basement into the wall system and collects behind the wall. In the Chicago tests, SWA found that allowing walls to dry by means other than diffusion through the insulation itself appeared adequate, and moisture diffusion into wall cavities did not cause problems.

In designing the Chicago study, SWA took great care to incorporate three framed-wall systems with vapor retarders of different classifications:

- class 1: polymer-encapsulated batts;

Detail at the Top of the Framed Wall Systems

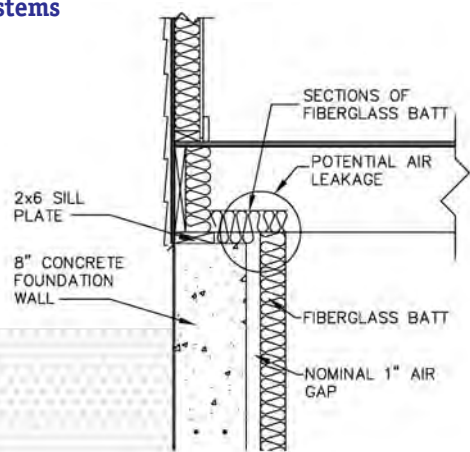


Figure 3. Air sealing at the top of the wall significantly reduced air movement behind the wall.

The improved drying capacity of the perforated poly-faced blankets was very slight, and SWA deduced that more diffusion occurred through the 6 inches of exposed concrete than through the insulation itself. In this Chicago home, this gap near the floor seemed much more important than the choice of perforated or nonperforated blankets.

The two rigid foam systems allow for wall drying (via diffusion) in very different ways. The foil facing on the polyisocyanurate boards is a class 1 vapor

wall; there is no gap near the floor. Moisture profiles behind and through this wall show that the system does allow the concrete to dry to the indoors adequately.

In neither of the two rigid foam systems did moisture diffusion into the wall system appear to cause problems.

Exterior Moisture

An unfortunate aspect of basements is that they are, essentially, large holes in the ground. This means that they are prone to collect water—primarily rain and groundwater. Good site planning, foundation drainage, and rainwater management are the best defenses against water leaking into foundations. Basement insulation systems do nothing to prevent this type of intrusion, but they should be able to recover from it.

One of the most important means of recovery from leaks is homeowner action; if homeowners take steps such as mopping up water, installing a dehumidifier, improving rain management, and so on, the damage caused by exterior moisture can be minimized. Homeowners can react to problems much more effectively if they know about them, and seeing bulk water is a great indication. Unfinished systems, where insulation stops 6 inches above the floor, give homeowners the best chance to see any such problems.

Rigid Foam Applications

The Cambridge study suggested that rigid foam or fiberglass blankets, installed properly, were among the best-performing, lowest-cost basement insulation systems. Foil-faced polyisocyanurate, because of its fire rating, can be installed without drywall and is the least expensive rigid foam system readily available. Based on the results of the research, SWA recommended foil-faced polyisocyanurate insulation to the builder of a Building America prototype home in Magna, Utah. The Community Development Corporation (CDC) of Utah also allowed SWA to monitor the performance of several different insulation systems.

Drying of Basement Walls Insulated with Polyisocyanurate

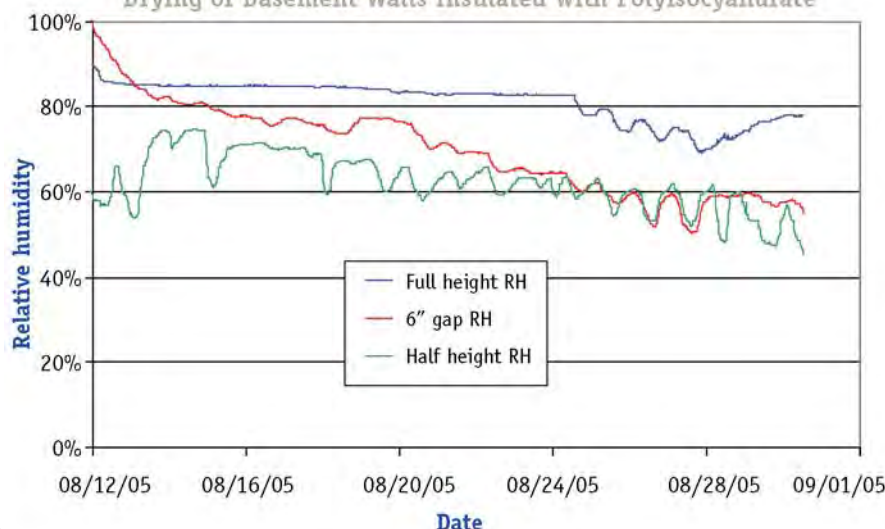


Figure 4. Relative humidity was measured behind the three polyisocyanurate insulation systems in Utah. Water was introduced behind all systems August 12.

- class 2: kraft-faced batts and drywall; and
- class 3: unfaced batts and drywall.

As noted above, there was no difference in the performance of these systems. Air movement dominated any and all moisture transport, and SWA feels that air sealing these framed-wall systems would require too much time, effort, and quality control to be expected of insulators or builders.

retarder; virtually no moisture can diffuse through the insulation itself. Moisture profiles behind the insulation show—as expected—that moisture in the foundation wall diffuses downward through the concrete and dries to the basement through the 6 inches of exposed concrete near the floor.

The drywalled EPS system acts as a class 3 vapor retarder with a permeance of approximately 2 perms. EPS and drywall are installed over the full

This smaller follow-up study was designed to answer more specific questions about the polyisocyanurate system:

- How much does the exposed-concrete gap at the bottom of the foundation wall matter?
- How big does this gap need to be?

To help answer these questions, SWA monitored the performance of three side-by-side systems installed in the Magna basement:

- half-height polyisocyanurate (installed over the top 4 feet of foundation wall);
- polyisocyanurate cut 6 inches above the slab floor; and
- full-height polyisocyanurate.

The Utah climate is very dry, so there was very little opportunity to evaluate moisture performance of basement walls under normal circumstances. SWA used this configuration to investigate the systems' responses to bulk water movement: SWA engineers introduced 2.25 liters of water at the top of each 8-inch-wide test section over a 24-hour period.

Though hardly a controlled, laboratory experiment, this test did provide additional insight into the systems' performances. The dry climate also enabled SWA to introduce bulk water without compromising long-term integrity; foam can be removed and the wall can dry quickly should high moisture levels persist.

It's clear that the full-height system retained moisture much more than the others (see Figure 4). The half-height system appeared to recover immediately, and the 6-inch system recovered completely in two weeks. The full-height insulation system had not recovered fully at the time of this writing. From this limited study, it appears that a small (6-inch) gap is adequate to allow bulk water to drain from the wall and to allow a modest amount of diffusion from the concrete.



Polyisocyanurate insulation was installed three ways, left to right: half-height, cut 6 inches from the floor, and full-height.

Some Recommendations

As with most research, there are still unanswered questions. However, the basement insulation study showed that systems with rigid foam attached directly to foundation walls resulted in better thermal and moisture-related performance.

As all framed-wall systems had some air space between the framed wall and the foundation wall, they all allowed basement air (and the moisture within it) to move behind the frame in contact with the basement wall. This resulted in a reduction in thermal efficacy of 10%–20%. Under some conditions, this mass transport also caused condensation on the cool foundation wall.

Performance of the fiberglass blanket systems was somewhere between that of the rigid foam and that of the framed-wall/batt systems. When detailed and installed correctly, fiberglass blankets certainly seem to be viable. Installation details should prevent convection through the system.

Based on this study and on related research, SWA's recommendations to builders who want to use interior basement insulation are as follows:

- Use rigid foam attached directly to the foundation wall. The simplest way to attach it is usually to use construction adhesive.

- If insulation is left exposed, use faced polyisocyanurate insulation (because of its fire rating).

- Stop insulation approximately 6 inches above the basement floor.

- If finishing the basement, still use rigid foam. EPS and XPS are also acceptable in such an installation, because the foam will be covered by drywall. Drywall can be attached to furring strips on the foam or on a light framed wall.

- If using fiberglass blankets, attach blankets securely and closely to the wall, so there will be no substantial air movement behind the system.



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